

Improved Grip Stability in Healthcare: Mixed Reality Assistance Devices for Degenerative and Age-Related Hand Conditions

Zhenhong Lei^{*†}

Architecture Department
Rhode Island School of Design
Providence, Rhode Island, US
brad.zh.lei@gmail.com

Xinjun Li^{*}

Information Science (Urban Tech), Cornell Tech
Cornell University
New York, New York, US
xl863@cornell.edu

Abstract

This study explores integrating Mixed Reality (MR) and Human-Computer Interaction (HCI) technologies to develop an advanced assistive device. The device improves grip stability for individuals suffering from Rheumatoid Arthritis (RA) and Osteoarthritis (OA). It preserves autonomy and improves overall well-being by combining MR with a servo motor locking mechanism to provide enhanced grip assistance. The concept incorporates digital overlays that deliver real-time guidance and feedback to aid physical manipulation and enhance rehabilitation, making it more immersive and tailored to individual needs. It outlines an iterative design approach that merges ergonomic design principles with sophisticated engineering, demonstrating MR's pivotal role in enhancing digital and physical interactions. Initial feedback from users has indicated significant enhancements in task performance and satisfaction, highlighting the device's capacity to promote independence and mitigate user discomfort. This study underscores MR's substantial potential to redefine the functionality of assistive devices, ultimately fostering greater autonomy and improved quality of life for individuals living with RA and OA.

CCS Concepts

• **Human-centered computing~Accessibility~Accessibility technologies** • **Human-centered computing~Human computer interaction (HCI)~Interaction paradigms~Mixed / augmented reality** • **Applied computing~Life and medical sciences~Health care information systems**

^{*}Authors contributed equally and shall both be considered first authors.

[†] Contact author

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

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ACM 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3678496>

Keywords

Augmented Reality, Mixed Reality, Assistive Device, Haptic Feedback, Geriatric Care, XR Healthcare, Diversity Equity and Inclusion (DEI), Hand Dexterity, Ergonomic Hand Motion

ACM Reference format:

Zhenhong Lei & Xinjun Li. 2024. Improved Grip Stability in Healthcare: Mixed Reality Assistance Devices for Degenerative and Age-Related Hand Conditions. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678496>

1 Introduction

The escalating prevalence of degenerative conditions such as Rheumatoid Arthritis (RA) and Osteoarthritis (OA), alongside a general decline in muscle strength due to aging, underscores the pressing need for sophisticated assistive technologies. These ailments severely compromise hand functionality, impeding daily activities and diminishing life quality. As the world's population ages, the incidence of these disorders is increasing, highlighting the critical need for effective and personalized solutions to improve grip stability and hand mobility. [1]

This paper broadens the scope of Mixed Reality (MR) applications from previous studies focused on older adults to encompass individuals with RA and OA, acknowledging that these conditions affect the aging population, necessitating a broader application of innovative assistive technologies to these groups.

Our MR-enhanced assistive device merges MR technology with a servo motor locking mechanism. This novel integration provides essential mechanical support alongside real-time, MR-driven instructional feedback, enhancing grip stability and encouraging active participation in rehabilitation activities. This user-centric design process demonstrates the potential of this technology to transform the landscape of assistive devices for those with grip-related disabilities.

2 Methodology

2.1 Comparative Analysis

Current interventions for managing hand functionality impairments due to Rheumatoid Arthritis (RA), Osteoarthritis (OA), and age-related decline typically involve mechanical assistive devices like customized gloves and grip aids. These devices facilitate object manipulation by enhancing the user's ability to hold and manipulate objects, directly addressing the physical limitations [2]. In contrast, digital tools, especially those utilizing Mixed Reality (MR), focus on rehabilitating motor skills through interactive and immersive exercises that simulate everyday activities [3].

However, a review of the existing literature reveals a significant gap in integrating these mechanical supports with comprehensive rehabilitative strategies. Typically, solutions have been bifurcated, providing either immediate mechanical support or focusing on long-term rehabilitative outcomes but not effectively combining both. Mechanical aids often offer prompt but rigid support, which may lack adaptability and fail to engage users fully. Meanwhile, MR platforms provide dynamic, engaging exercises but may not offer immediate physical support [4, 5].

Our innovative MR-enhanced device addresses this gap by integrating real-time physical support with interactive digital rehabilitation. This dual approach provides immediate mechanical assistance and promotes neurological adaptations through engaging, personalized exercises, offering a comprehensive solution to hand functionality issues. This system is designed to enhance immediate grip stability and contribute to long-term improvements in hand mobility and strength, thereby significantly enhancing user independence and quality of life [6]. This integrative strategy represents a significant advancement in the field, introducing a new paradigm in developing assistive devices that meet the complex needs of individuals with degenerative and age-related conditions.

2.2 Research and Development Process with Comparative Focus

2.2.1 Research Phase. The research initiative began with an exhaustive review of current therapeutic approaches to managing Rheumatoid Arthritis (RA) and Osteoarthritis (OA). This review identified gaps in the effectiveness of existing physical aids and virtual rehabilitation techniques. Consultations with clinical experts helped tailor the development approach to meet these patient groups' specific needs.

2.2.2 Ideation Phase. The concept of a mixed-reality device was developed. This device integrates physical joint support with interactive augmented reality (AR) rehabilitation. The idea was continuously refined through iterative feedback from healthcare professionals and potential users, ensuring its relevance and practical utility.

2.2.3 Design & Development Phase. This phase focused on enhancing traditional support mechanisms by incorporating a robust AR component to facilitate effective rehabilitation exercises.

2.2.4 Testing Phase. The prototype underwent rigorous testing against standard RA and OA treatment protocols to assess

the efficacy of the physical support and user engagement with the AR system.

2.2.5 Refinement Phase. The insights gained from the initial testing were crucial in refining the device, which was intended to verify its effectiveness and confirm its benefits in offering a comprehensive treatment strategy for RA and OA.

From the initial concept to the final validation, the development process reflects our commitment to creating a tailored solution that addresses both the immediate and evolving needs of RA and OA patients. This process exemplifies our dedication to merging advanced technology with a user-centered design ethos, underscoring the comprehensive approach taken in this research.

3 Design Process

3.1 Ideation

In our endeavor to tackle the complexities associated with Rheumatoid Arthritis (RA) and Osteoarthritis (OA), our approach to designing an assistive device was guided by a comprehensive methodology drawing from multiple disciplines. This methodology integrates principles from biomechanics, ergonomics, and user-centered design. Our prototype was meticulously crafted to accurately mimic the natural mechanics of hand movements, with a specific focus on facilitating the fundamental grip necessary for everyday activities. By prioritizing biomechanical accuracy, the device's structure was intentionally fashioned to conform to the innate curvature of the hand at rest, thereby enhancing comfort and reducing the risk of strain. (Figure 1)

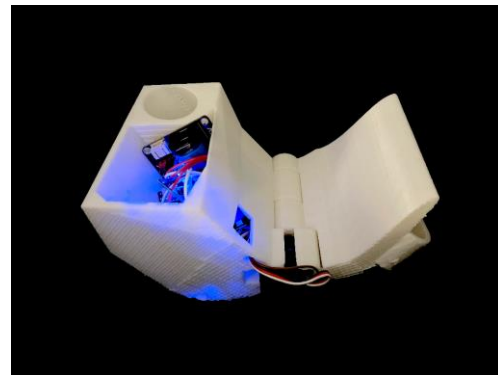


Figure 1: Device Mock-up.

During the ideation phase, multiple design iterations were undertaken to tackle specific challenges to improve user-friendliness and ergonomic functionality. This iterative method allowed for a thorough exploration of the physical characteristics of the intended device, facilitating a comparative assessment of the strengths and weaknesses of each iteration. Furthermore, this approach facilitated swift prototyping, which is crucial for upholding stringent design quality benchmarks during development. (Figure 2)

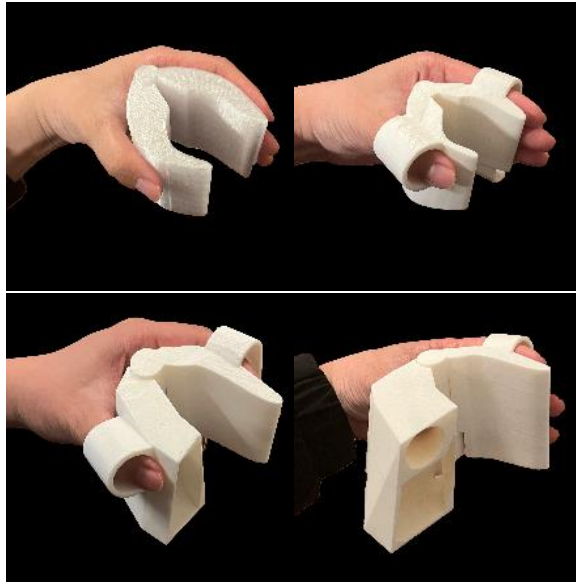


Figure 2: Iteration Process.

3.2 Physical Device Prototyping

The inclusion of a servo motor within the device aims to optimize the functionality of hand opening and closing actions, which are pivotal for sustaining muscle endurance during activities requiring prolonged grip, such as holding a book or gripping handles securely. This attribute holds significance for individuals grappling with Rheumatoid Arthritis (RA) and Osteoarthritis (OA), conditions characterized by joint stiffness and discomfort, often compromising grip strength and fine motor skills. By integrating the servo motor, the design achieves a synergistic fusion of mechanical assistance and inherent hand motions, effectively empowering users to navigate daily tasks with enhanced independence. (Figure 3)



Figure 3: The user holding a book with the device's help.

The design concept for enabling the transition between the hand's open and closed states involves a dual-process approach. Initially, a user-friendly trigger mechanism is engineered to initiate the grasping action, effectively counteracting reduced grip strength.

Subsequently, a secondary mechanism is activated by a simple ergonomic command to facilitate the release action. This command disengages the servo motor, allowing the hand to return to its natural open position.

Our device utilizes a dual-component system connected by a hinge, incorporating a feedback servo motor located in the thumb to mimic natural hand movements. This design aids individuals with Rheumatoid Arthritis (RA) or Osteoarthritis (OA) in gripping and releasing actions with enhanced stability and reduced strain. Additionally, a gearing system adjusts motor stiffness, optimizing grip strength for specific tasks. User-initiated commands unlock the motor, enabling a smooth and controlled release, thereby returning the hand to its open position (Figure 4).

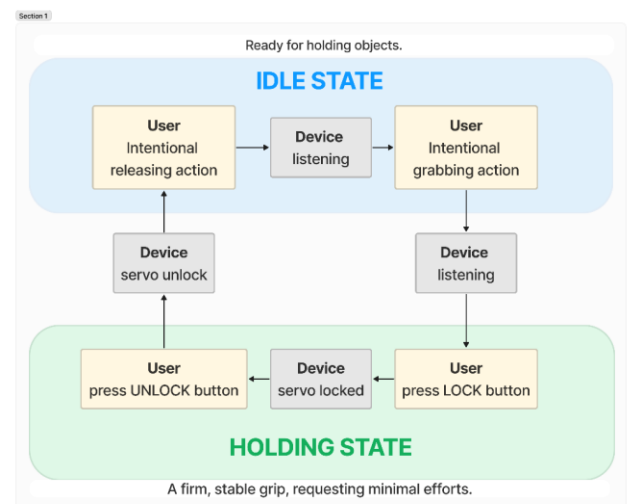


Figure 4: Diagram of the device functionality and the flow of information.

This functionality emulates organic hand gestures and integrates sophisticated engineering concepts to amplify everyday utility and self-sufficiency among older adult demographics.

3.3 AR System Development

In our pursuit of improving the quality of care for elderly individuals affected by Rheumatoid Arthritis (RA) and Osteoarthritis (OA) through technological advancements, we have devised an Augmented Reality (AR) system. Leveraging Unity, it harnesses a smartphone's camera to identify everyday objects in the user's environment. Our application furnishes intuitive AR overlays to assist users with impaired dexterity in daily tasks.

This system's functionality extends to tasks such as handling books and providing visual prompts for optimal manipulation. A feedback loop is established by mirroring the user's actions on the smartphone screen, reinforcing correct motion patterns, and improving muscle memory. Upon activation of the "Grab suggestion" feature, the application initiates a comprehensive analysis of the object using successive camera frames. This

advanced analysis generates recommendations for the most effective grasp of the object.

As the user attempts to grasp the object, their movements are instantly mirrored on the smartphone display, facilitating real-time feedback and guidance (Figure 5). This innovative approach holds promise in enhancing the independence and quality of life for elderly individuals with RA and OA by empowering them with assistive technology tailored to their specific needs.

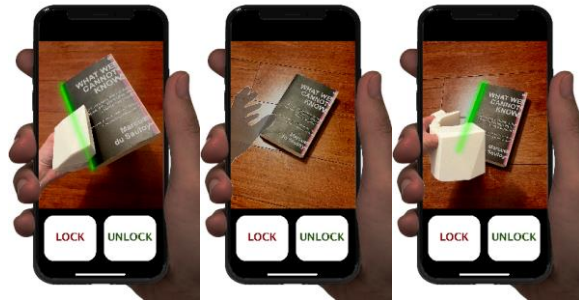


Figure 5: Smartphone AR interface.

This real-time representation is facilitated by a gyroscope integrated into the device, transmitting precise movement data to the application. The application incorporates an adaptive UI/UX framework, dynamically adjusting interface elements such as button sizes and touch sensitivity settings based on real-time interaction analysis. These adjustments cater to users with Rheumatoid Arthritis (RA) and Osteoarthritis (OA), mitigating joint stiffness and pain during interaction, thereby enhancing application usability and efficiency. Additionally, integrating haptic feedback alongside visual cues from the Augmented Reality (AR) application aims to augment tactile user interactions, closing the sensory feedback loop and enriching the overall user experience.

The design also explores the practicality of stabilizing accessories like gloves or braces equipped with smartphone holders. However, these were found to be less suitable for elderly users, often hindering natural hand movements and increasing complexity, particularly for those with severe RA or OA or limited mobility.

3.4 User Testing

The preliminary qualitative evaluation of the device involved a focus group of individuals diagnosed with Rheumatoid Arthritis (RA) and Osteoarthritis (OA). Valuable feedback from this cohort informed iterative improvements to the device's functionality and user comfort. Participants reported increased ease in daily activities and decreased discomfort, emphasizing the device's ergonomic design and user-friendly interface. While the findings are based on initial qualitative feedback and lack comprehensive quantitative analysis, they underscore the device's potential and identify areas for further refinement. The augmented reality (AR) feature and telehealth functionalities elicited positive responses, indicating the need for more extensive studies to thoroughly assess the device's effectiveness and optimize its design for broader user adoption.

4 Discussion

Our exploration of a mixed reality (MR) device tailored for Rheumatoid Arthritis (RA) and Osteoarthritis (OA) integrates assistance for ergonomic hand motions with augmented reality rehabilitation. This combination has shown promising initial results, highlighting the necessity for further empirical investigation. This integration of physical and virtual rehabilitation aligns with contemporary trends in MR healthcare, emphasizing improvements in user experience and adaptability in subsequent device iterations. However, the complexity of the augmented reality system poses challenges, highlighting the importance of developing more user-friendly interfaces tailored specifically for RA and OA populations. Such feedback is crucial for iteratively refining the device to meet RA and OA patients' ergonomic and cognitive requirements. Future iterations of the device should prioritize enhancing user engagement and accessibility by tailoring the technology to accommodate the diverse needs and preferences of RA and OA patients.

5 Conclusion

In this study, we investigate the fusion of Mixed Reality (MR) and Human-Computer Interaction (HCI) for developing therapeutic interventions targeting Rheumatoid Arthritis (RA) and Osteoarthritis (OA). Our device combines ergonomic hand motion support with augmented reality technology, tailored to address these conditions' physical and perceptual challenges. Designed to adapt to diverse hand sizes and joint mobility constraints, it aims to enhance accessibility.

Analysis of empirical data from participants reveals notable improvements in motor control and autonomy, affirming MR's potential to enhance quality of life significantly. Our future endeavors will focus on refining user interface design and deepening the integration of MR and HCI to bolster the efficacy of rehabilitation, catering to a broader user demographic. This underscores MR's substantial promise in creating inclusive and effective therapeutic approaches for RA and OA.

References

- [1] A. Bamford and E. J. Henderson, "Parkinson's disease in older people," *Medicine*, vol. 49, no. 1, pp. 56–61, Jan. 2021, doi: <https://doi.org/10.1016/j.jmpmed.2020.10.008>.
- [2] M. Schwalbe, S. Satz, R. Miceli, H. Hu, and A. Manelis, "Hand Dexterity Is Associated with the Ability to Resolve Perceptual and Cognitive Interference in Older Adults: Pilot Study," *Geriatrics*, vol. 8, no. 2, p. 31, Apr. 2023, doi: <https://doi.org/10.3390/geriatrics8020031>.
- [3] M. F. Pereira, C. Prahm, J. Kolbenschlag, E. Oliveira, and N. F. Rodrigues, "Application of AR and VR in hand rehabilitation: A systematic review," *Journal of Biomedical Informatics*, vol. 111, p. 103584, Nov. 2020, doi: <https://doi.org/10.1016/j.jbi.2020.103584>.
- [4] G. Mois et al., "Perceptions from People Aging with a Mobility Impairment towards using Tele-Technology for Exercise," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 63, no. 1, pp. 11–15, Nov. 2019.
- [5] A. Kolsanov, S. Chaplygin, S. Rovnov, and A. Ivaschenko, "Mixed Reality Application for Hand Motor Skills Rehabilitation," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 4, 2020, doi: <https://doi.org/10.14569/ijacsa.2020.0110408>.

- [6] V. Dineshkumar, D. R. J. Dolly, D. J. Jagannath, and J. D. Peter, “Assistive Methodologies for Parkinson’s Disease Tremor Management—A Health Opinion,” *Frontiers in Public Health*, vol. 10, no. 850805, Apr. 2022, doi: <https://doi.org/10.3389/fpubh.2022.850805>.
- [7] Y. Gu et al., “A Review of Hand Function Rehabilitation Systems Based on Hand Motion Recognition Devices and Artificial Intelligence,” *Brain Sciences*, vol. 12, no. 8, 2022, doi: <https://doi.org/10.3390/brainsci12081079>.
- [8] I. Gonsher and Z. Lei, “Prototype of a force-feedback tool for mixed reality applications,” *Interactions*, vol. 29, no. 2, pp. 23–23, Feb. 2022, doi: <https://doi.org/10.1145/3514206>.
- [9] X. Li and Z. Lei, “Ergonomic Hand Motion Assistance and AR Rehabilitation: Bridging the Gap in Tremor Disorder Therapies through Mixed Reality,” 2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), Orlando, FL, 2024.